

Rajarata University of Sri Lanka

Department of ICT
ICT 1209- Web Technologies
Mini Project Final Report

Project Title:
Digital Recipe Book
(Veganfood)

Theme	Digital Recipe Book
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Group member 2	H.T.T.D.Nishantha (index number 2761)
GitHub Repository	https://github.com/TharushaDasun23/Digital-Recipe-Book.git
Date	2026/09/06

1.project overview

1.1 purpose

The project is named '**VeganFood**' specifically because it focuses exclusively on **100% plantbased and vegan recipes**. The platform provides a dedicated space for users looking for strictly vegan meals without any animal-based ingredients.

1.2 Target Audience

Home cooks, students, food bloggers, and anyone seeking structured, accessible cooking recipes online.

1.3 key Features

- User registration and login with hashed passwords (register.php, login.php)
- Session-based authentication with session_regenerate_id() on login and clean logout
- Real-time recipe search by title/category (search.js)
- Category-based recipe filtering without a page reload (filter.js)
- User-submitted recipes stored in the MySQL database
- Contact form with server-side storage of messages (contact.php)
- Client-side form validation for the search bar and contact form (validation.js)
- Smooth scroll-to-top control (scroll.js)
- Hover/lift animation on recipe cards (hover.js)
- Dark/Light theme toggle and page-transition animations (theme.js, transition.js)

2. System Design

2.1 Database Schema

Database name: **veganfood_db**

The database consists of three tables as required by the project specification:

Users

Field	Type	Description
id	INT, AUTO_INCREMENT, PK	Unique user identifier
username	VARCHAR(50)	Display name / login name
email	VARCHAR(100), UNIQUE	User email address
password	VARCHAR(255)	Hashed password (PASSWORD_BCRYPT)
created_at	TIMESTAMP	Account creation time
role	VARCHAR(50)	User role (admin/user)

messages

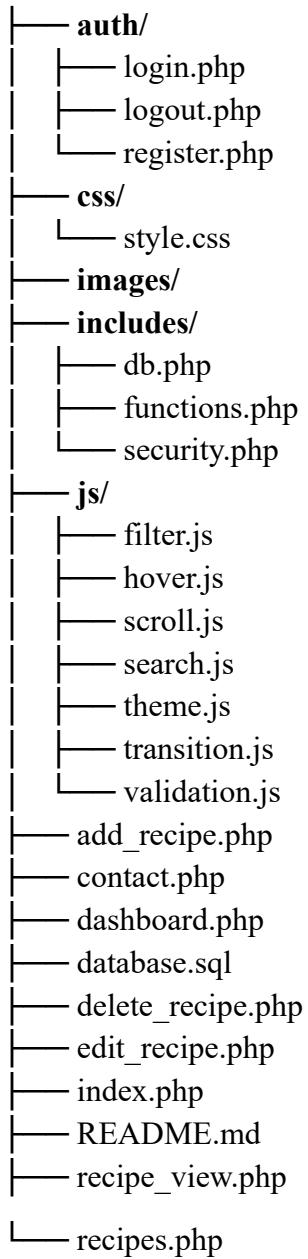
Field	Type	Description
id	INT, AUTO_INCREMENT, PK	Unique message identifier
user id	INT, FK	Sender's id
name	VARCHAR(100)	Sender's name
email	VARCHAR(100)	Sender's email address
message	TEXT	Message content
Created at	TIMESTAMP	Submission time

recipes (theme-specific table)

Field	Type	Description
id	INT, AUTO_INCREMENT, PK	Unique recipe identifier
user id	INT, FK	ID of the user who created the recipe
title	VARCHAR(255)	Name or title of the recipe
cuisine	VARCHAR(100)	Cuisine type or category (e.g., Sri Lankan, Italian)
Cooking time	INT	Preparation/cooking time in minutes
servings	INT	Number of servings
Image url	VARCHAR(2048)	URL link to the recipe image
ingredients	TEXT	List of required ingredients
instruction	TEXT	Step-by-step cooking instructions
Created at	TIMESTAMP	Record creation date and time

2.2 Folder Structure

veganfood/

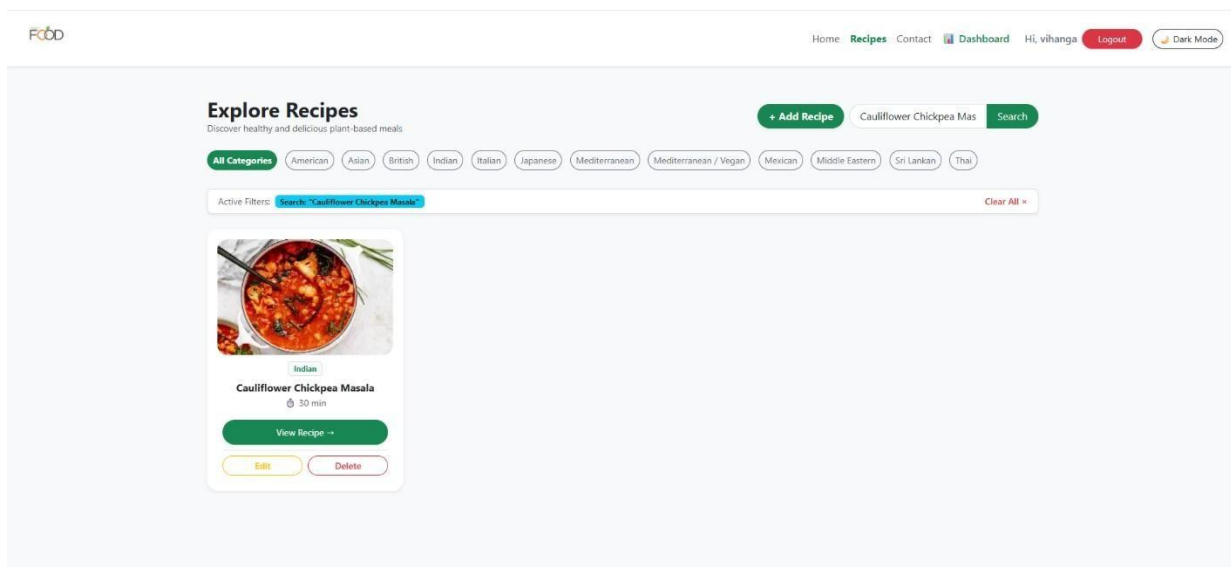


3. Implementation Details

This section describes each major feature implemented, grouped by the JavaScript, PHP, and database functionality required by the project brief.

3.1 Dynamic Content Updates

Implemented in search.js and filter.js. search.js listens for keyup events on the search input and filters the visible recipe cards in real time by matching the query against the recipe title and category, without reloading the page. filter.js provides category filter buttons; clicking a button prevents the default link navigation, updates the active button styling, and shows or hides recipe cards based on the data-category attribute, re-triggering a staggered fade-in animation for the results that remain visible.



search/filter

3.2 Form Validation

Implemented in validation.js. The search form checks that the query is non-empty and contains at least one alphanumeric character before submission, and cross-checks it against a list of known recipe names/categories, showing an inline error message via the input's placeholder and an is-invalid class when no match is found. The contact form (contact.php) is validated on the client side for name length, a valid email pattern, subject length, and a minimum message length, with invalid fields marked using Bootstrap's is-invalid class. This client-side validation runs alongside, not instead of, server-side validation in PHP.

Contact Us

Name
vihanga lakshan

Email
Test

Subject
Please include an '@' in the email address. 'Test' is missing an '@'.
rescipes

Message

[Send Message](#)

Copyright © 2026 VeganFood. All rights reserved.
contact us

3.3 Smooth Scrolling

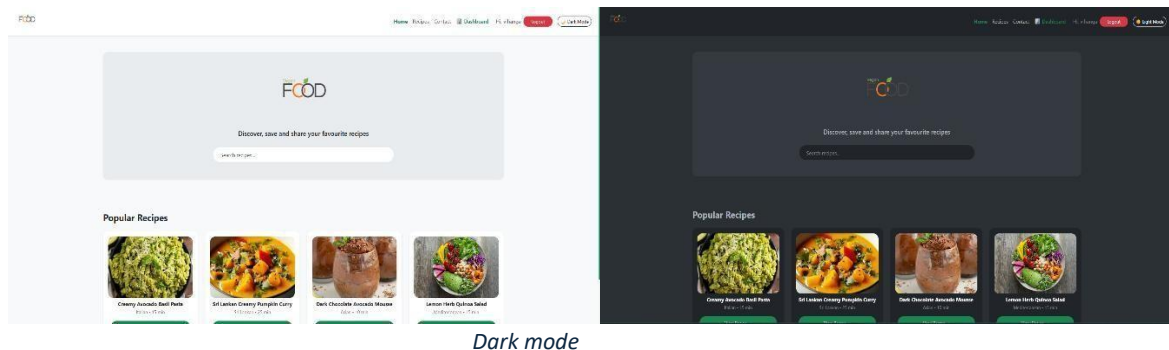
Implemented in scroll.js. A floating "scroll to top" button is hidden by default and revealed once the user scrolls more than 200px down the page. Clicking the button calls window.scrollTo with behavior: "smooth" to animate the return to the top of the page.

3.4 Event Handling

Implemented in `hover.js`. Each recipe card listens for `mouseenter` and `mouseleave` events; on hover, the card lifts by 6px using a CSS transform with a 0.3s transition, giving clear visual feedback that the card is interactive.

3.5 Custom Animations

Implemented in `theme.js` and `transition.js`. Recipe cards use a staggered entrance animation, with each card's animation-delay incremented based on its position in the grid so cards appear in sequence rather than all at once. `theme.js` also implements a dark/light mode toggle that stores the user's preference in `localStorage` and updates the Bootstrap `data-bs-theme` attribute and button label accordingly. `transition.js` adds a fade-style page exit animation: internal link clicks are intercepted, a page-exiting class is applied, and navigation is delayed briefly so the transition can play before the new page loads.



3.6 User Authentication (PHP & MySQL)

register.php collects a username, email, and password, hashes the password with password_hash() using PASSWORD_BCRYPT, and inserts the new user into the users table using a prepared statement. login.php verifies the submitted credentials with password_verify(), starts a session on success, calls session_regenerate_id() to protect against session fixation, and redirects the user to dashboard.php. logout.php destroys the session cleanly and redirects to the login page.



Register

Register

Username:

Email Address:

Password:

REGISTER

Already have an account? [Login Here](#)
[Back to Home](#)



Login

Welcome Back!

Email Address:

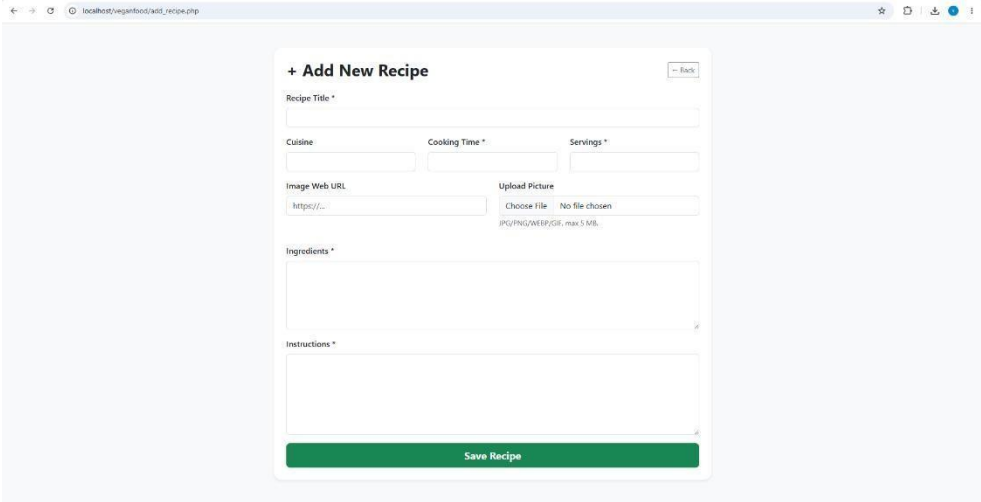
Password:

LOGIN

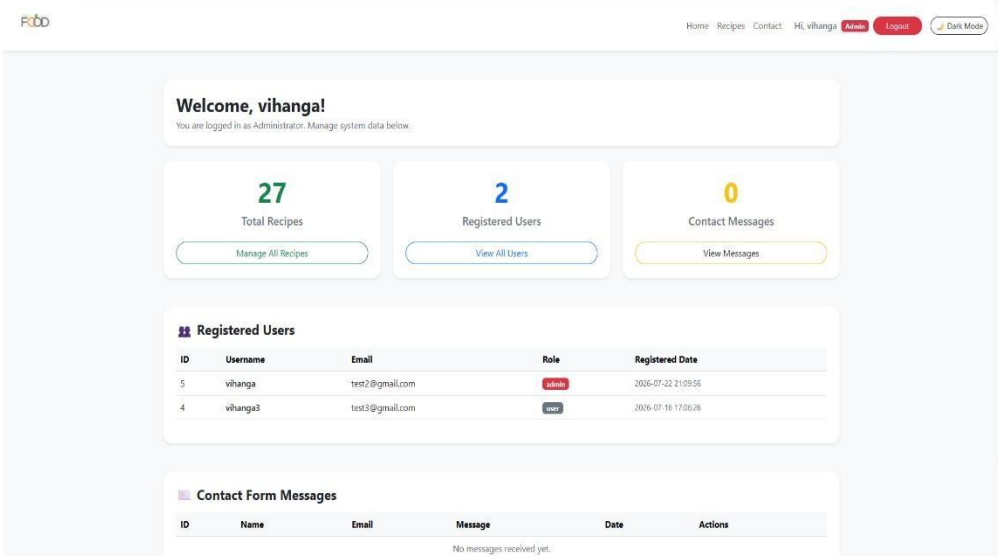
New to VegerFood? [Create an Account](#)
[Back to Home](#)

3.7 Recipe Submission and Contact Form (PHP & MySQL)

Authenticated users can submit a new recipe with a title, category, ingredients, instructions, and an image; the data is inserted into the recipes table with the submitter's user_id via a prepared statement. The contact.php page stores each submitted enquiry in the messages table and displays a confirmation message to the user on success. Both forms use PDO/MySQLi prepared statements throughout to prevent SQL injection.



Add New Recipe



Welcome, vihanga!
You are logged in as Administrator. Manage system data below.

27
Total Recipes
[Manage All Recipes](#)

2
Registered Users
[View All Users](#)

0
Contact Messages
[View Messages](#)

Registered Users

ID	Username	Email	Role	Registered Date
5	vihanga	test2@gmail.com	Admin	2026-07-22 21:09:56
4	vihanga3	test3@gmail.com	User	2026-07-16 17:56:28

Contact Form Messages

ID	Name	Email	Message	Date	Actions
No messages received yet.					

Dashboard

4. Challenges and Solutions

Challenge	Solution
The database connection was not establishing correctly between PHP and MySQL.	Checked the credentials and host settings in includes/db.php against the local XAMPP MySQL configuration, confirmed the veganfood_db database existed in phpMyAdmin, and retested the connection with a simple query before wiring it into the rest of the application.
Uploaded recipe images were not displaying because of incorrect file paths.	Standardised the images/ folder location relative to the project root and updated all image src attributes and stored database paths to use consistent relative paths, then re-tested on each page.
Some JavaScript features were not functioning as expected on certain pages.	Traced the issue to scripts running before the target elements existed in the DOM and to missing element IDs on some pages; wrapped the logic in DOMContentLoaded listeners and added existence checks before attaching event handlers, then re-tested each script individually.
Encountered configuration issues while setting up the local server with XAMPP.	Resolved port conflicts on Apache/MySQL, restarted the XAMPP control panel services, and verified the project folder (veganfood) was correctly placed inside htdocs so the application could be accessed via localhost (http://localhost/veganfood/).

5. Individual Contribution

Both group members worked collaboratively across the frontend, backend, and database components of the project, and jointly debugged and fixed errors in each other's code throughout development. A summary of contributions is given below; commit history on GitHub reflects the collaborative nature of the work.

Member	Contribution
T.M.V.L. Weerasekara (Index number 2798) ITT/2024/113	Contributed to frontend layout and styling, JavaScript interactivity, and participated in debugging and fixing errors across the codebase alongside the second member.
H.T.T.D. Nishantha (Index number 2761) ITT/2024/076	Contributed to PHP backend logic, database integration, and participated in debugging and fixing errors.

6. Setup Instructions

Follow the steps below to run the project locally using XAMPP:

- Install XAMPP and start the Apache and MySQL services from the XAMPP Control Panel.
- Copy the project folder into the htdocs directory(e.g.c:\xampp\htdocs\veganfood).
- Open phpMyAdmin (<http://localhost/phpmyadmin>) and create a new database named `veganfood_db`
- Import the provided database.sql file into the **veganfood_db** database using the Import tab in phpMyAdmin.
- Update the database credentials (host, username, password, database name) in `includes/db.php` if they differ from the default XAMPP settings.
- Open a browser and navigate to <http://localhost/veganfood/index.php> to launch the application.
- Register a new account, log in, and test the recipe search, filter, submission, and contact form features.

7. References

- MySQL Documentation – <https://dev.mysql.com>
- Bootstrap 5 Documentation – <https://getbootstrap.com>
- Mozilla Developer Network (MDN) – <https://developer.mozilla.org>
- W3Schools – PHP and MySQL Tutorials – <https://www.w3schools.com>